$Ti_3C_2T_x$ MXene — Accordion-Structured Multilayer Powder (HF-Etched)

Representative Technical Data Sheet | WM-MX-TI3C2TX-ACC-ML | TI3C2TX-ACC-ML-TDS-001 | Revision 1

Issue date: November 7, 2026

1. Product identification

$Ti_3C_2T_x$ MXene; Ti_3AlC_2 precursor; multilayer powder prepared by HF etching.

2. Product overview

Accordion-like stacked lamellae provide a morphology option for powder processing and interfacial research. This grade is distinct from the LiF/HCl-derived clay-like multilayer grade (representative conductivity 1000 ± 500 S/cm). The HF grade is reported at 10–100 S/cm. These are not controlled comparative measurements and do not establish superior electrochemical performance.

3. Representative technical properties

All values below are supplier-reported representative information, not guaranteed release specifications.

Qualitative supplier description; no quantitative water-solubility measurement method or result was provided.

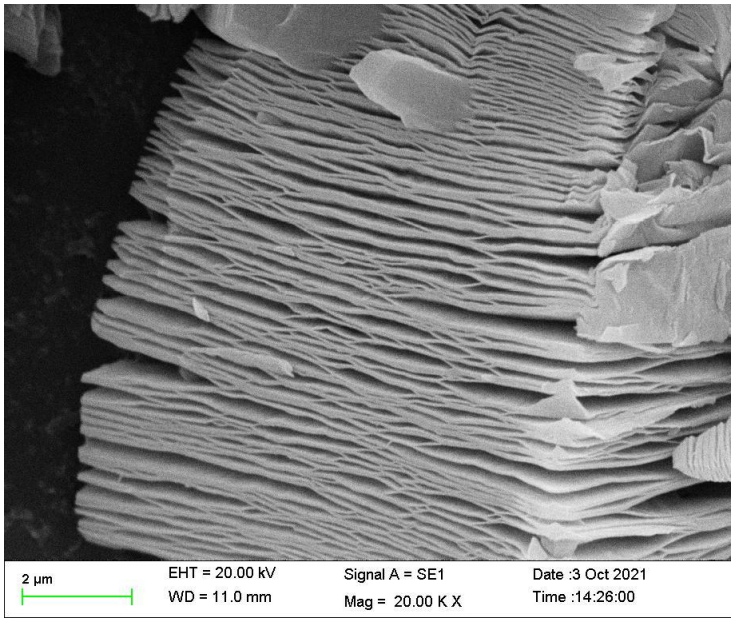
Property	Representative information
Material	$Ti_3C_2T_x$ MXene
Precursor	Ti_3AlC_2
Product form	Multilayer powder
Morphology	Accordion-like stacked lamellae
Preparation	HF etching
Appearance	Dark-purple powder
Representative conductivity	10–100 S/cm
Representative specific surface area	15 ± 5 m^2/g; method not specified
Surface terminations	Mixed –O, –OH, –F and –Cl
Water solubility	Poor (supplier-reported, qualitative)
Storage	Sealed and dry; preferably vacuum or inert atmosphere

4. Available technical grades

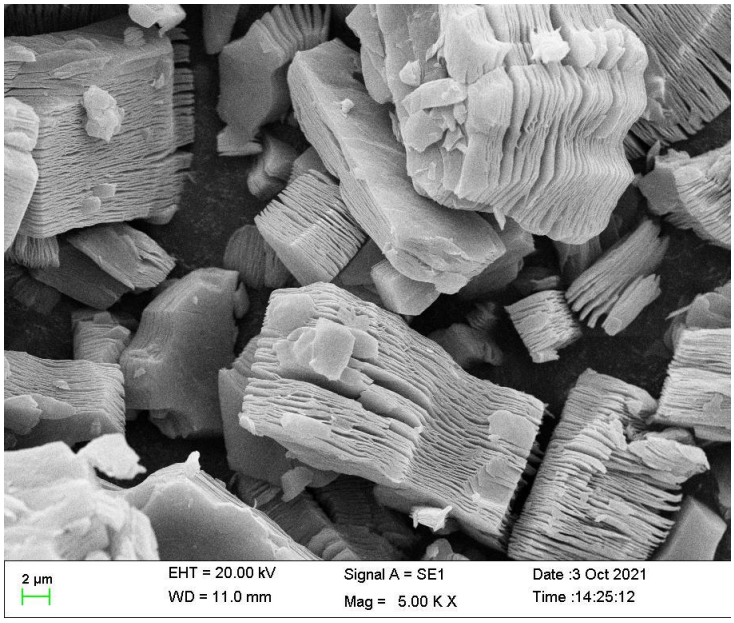
Grade A: −200 mesh precursor; nominal post-etching size 2–50 μm. Grade B: −400 mesh precursor; nominal post-etching size 2–20 μm. These are technical options, not guaranteed particle-size distributions or approved ordering packages.

5. Representative characterization

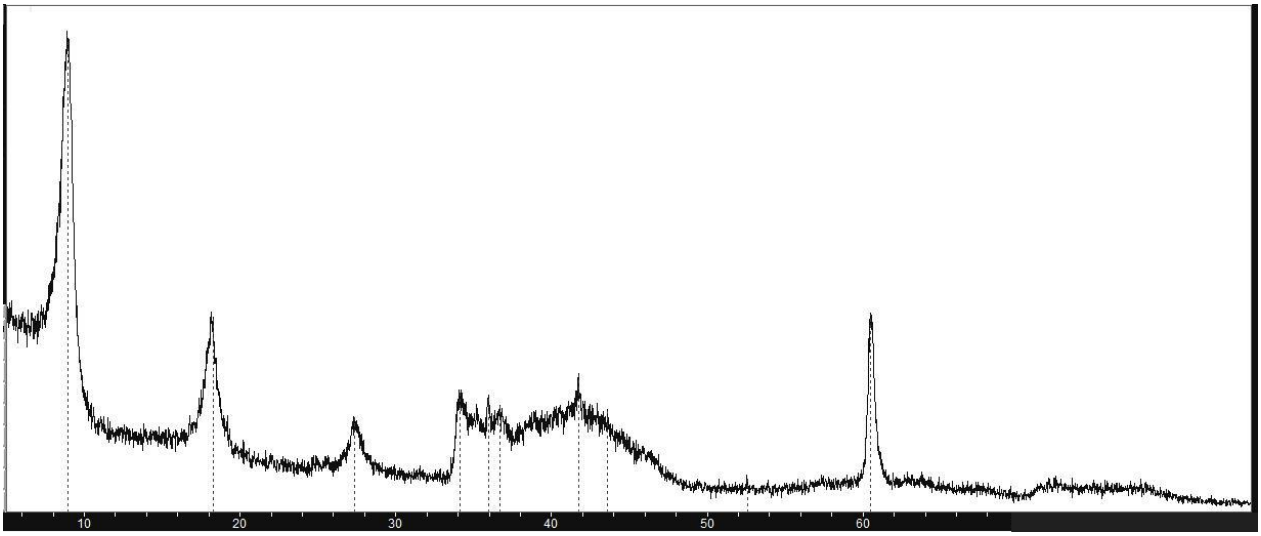
SEM field 1 - Representative characterization; not lot-specific.



SEM field 2 - Representative characterization; not lot-specific.



XRD - Representative supplier XRD pattern. Measurement conditions and axis definitions were not provided. Not lot-specific.



6. Storage and handling

Keep sealed and dry, preferably under vacuum or inert atmosphere. Minimize air and moisture exposure. Use appropriate laboratory controls for fine powders and consult the applicable SDS.

7. Technical disclaimer

This document provides representative technical information for research and evaluation. Values are not lot-specific release specifications and may vary with manufacturing conditions, preparation and measurement methods. Current-lot documentation may be requested where available.

Characterization is representative third-party reference data; it is not represented as testing performed by Winigen. No lot-specific COA is fabricated or supplied by this document.